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COURSE CODE: 24SDCS02

FULL STACK APPLICATION DEVELOPMENT

#SKILL - 4: Spring Dependency Injection – Constructor & Setter Injection

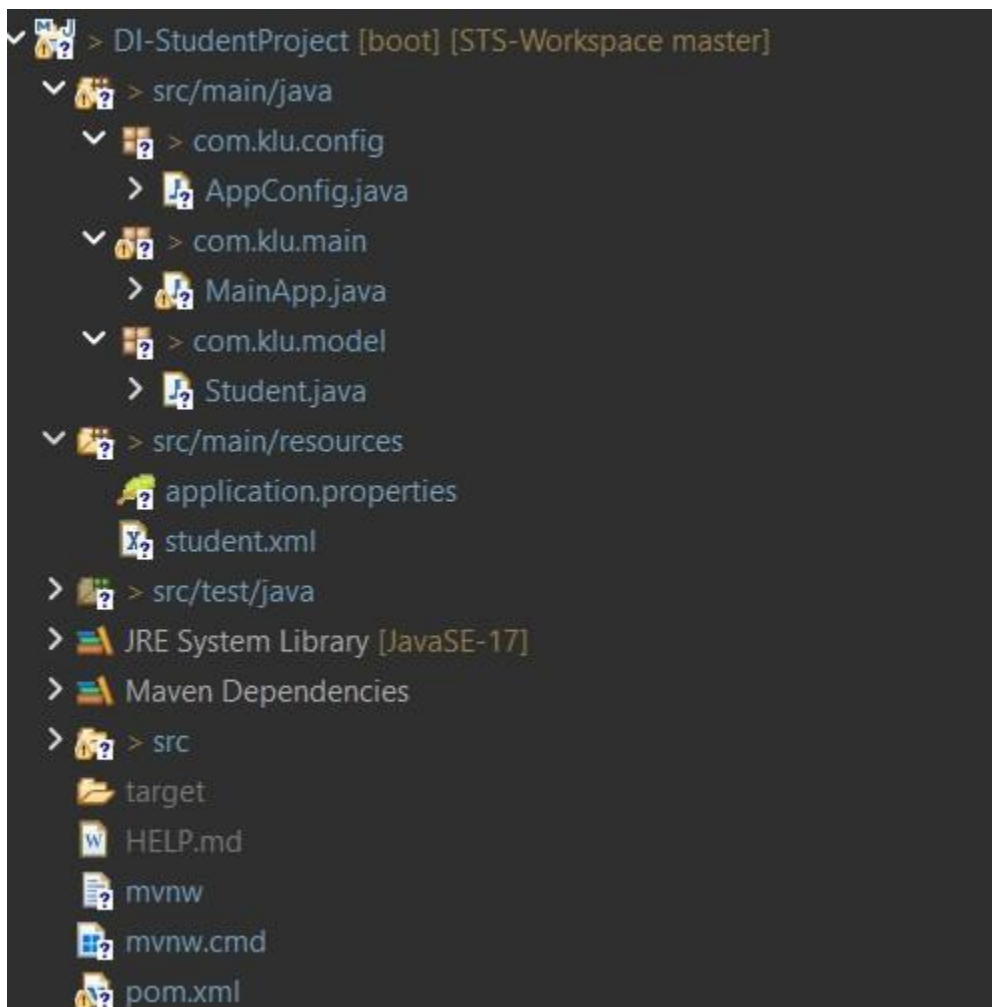
Prerequisites:

- Basic knowledge of Java
- General Idea on Spring Framework
- Basic Idea on DI & IoC

Tasks:

1. Create Student class with fields: studentId, name, course, year.
2. Implement constructor injection.
3. Implement setter injection.
4. Configure using XML and Annotations.
5. Load Spring IoC container and retrieve bean.

Structure:



AppConfig.java:

```
package com.klu.config;
```

```
import org.springframework.context.annotation.ComponentScan;
```

```
import org.springframework.context.annotation.Configuration;
```

```
import org.springframework.context.annotation.ImportResource;
```

```
@Configuration
@ComponentScan(basePackages = "com.klu.model")
@ImportResource("classpath:student.xml") // IMPORT XML INTO JAVA CONFIG
public class AppConfig {
}
```

MainApp.java:

```
package com.klu.main;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.klu.config.AppConfig;
import com.klu.model.Student;

public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context =
            new AnnotationConfigApplicationContext(AppConfig.class);

        // Annotation-based bean
        Student s1 = (Student) context.getBean("annoStudent");
        s1.display();
    }
}
```

```

        System.out.println(" ----- ");

        // XML-based bean
        Student s2 = (Student) context.getBean("xmlStudent");
        s2.display();
    }
}

```

Stident.java:

```

package com.klu.model;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

@Component("annoStudent")
public class Student {

    private int studentId;

    private String name;

    private String course;

    private int year;

    // Constructor Injection
    public Student(@Value("101") int studentId,
                  @Value("Saifulla") String name,
                  @Value("B.Tech") String course,
                  @Value("2") int year) {

```

```
    this.studentId = studentId;

    this.name = name;

    this.course = course;

    this.year = year;
}
```

```
// Setter Injection
```

```
@Value("FSAD")

public void setCourse(String course) {

    this.course = course;
}
```

```
@Value("2")

public void setYear(int year) {

    this.year = year;
}
```

```
public void display() {

    System.out.println("Student Details:");

    System.out.println("Student ID : " + studentId);

    System.out.println("Name      : " + name);

    System.out.println("Course   : " + course);

    System.out.println("Year      : " + year);

}
}
```

Student.xml:

```
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="xmlStudent" class="com.klu.model.Student">

        <constructor-arg value="101"/>
        <constructor-arg value="Saifulla"/>
        <constructor-arg value="Java"/>
        <constructor-arg value="1"/>

        <property name="course" value="FSAD"/>
        <property name="year" value="2"/>
    </bean>

</beans>
```

OUTPUT:

```
Student ID: 101
Name: Saifulla
Course: B.Tech
Year: 2
```

```
[ INFO] ..s.o.j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFa
```

VIVA QUESTIONS:

1. What is HQL and how does it differ from SQL?

DI is a technique where Spring provides the required objects to a class instead of the class creating them itself.

2. Why do we use sorting in HQL?

Constructor DI injects dependencies through the constructor, while Setter DI injects them using setter methods.

3. What is pagination and where is it used?

The IoC container creates, manages, and injects objects (beans) in a Spring application.

4. Name any three aggregate functions supported in HQL.

DI improves loose coupling, easier testing, and better code maintainability.

5. What is the difference between HQL and HCQL (Hibernate Criteria Query Language)?

Yes, both constructor and setter injection can be used together in the same class.

Git Hub Link : <https://github.com/saifulladm007>